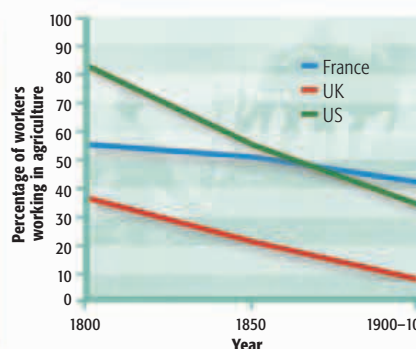
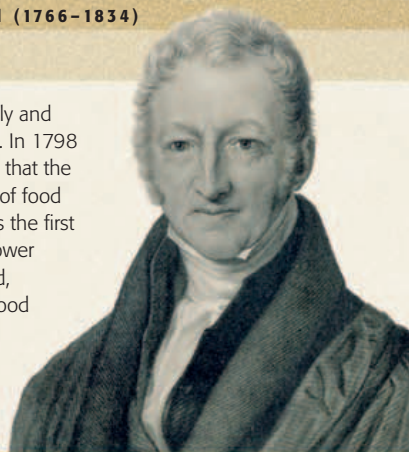


BRITISH ECONOMIST AND HISTORIAN (1766–1834)

THOMAS MALTHUS

Thomas Malthus was born into a wealthy family and enrolled at Jesus College, Cambridge, in 1784. In 1798 he published his *Essay on Population*, arguing that the growth in population was far outstripping that of food production. Although derided by many, he was the first to write about the economic situation of the lower classes. Malthus believed that if left unchecked, population would always exceed the level of food available. He saw starvation and disease as “positive checks” on population growth and voluntary abstinence as a form of birth control as “preventative checks.”



Agricultural labor

The technological innovations of the 19th century changed the way the labor force was deployed. As food production increased, workers moved away from agriculture, into industry.

of other farmers. In the US, the mechanical reaper, invented by Cyrus Hall McCormick in 1831, further reduced the need for farm labor. Steam power invaded the countryside with the introduction of the Fowler steam plow in the 1850s.

In the second half of the 18th century better livestock was introduced with selective breeding, by British agriculturalists Robert Bakewell and Thomas Coke. This increased profit and food supplies. In the mid-19th century, the British scientists Joseph Gilbert and John Lawes demonstrated that plants required nitrogen and other nutrients, improving techniques for farming crops.

Around the world

As Europeans explored the globe, they carried with them their farming methods and crops (see pp.232–33).



Food preservation

These food cans date from the Boer War (1899–1902). During wars, demand for canned food increased. At home, it was a status symbol among the middle classes.

In 1788, the first European settlers arrived in Australia, and influenced early practices in wool production. In the US thousands had moved west in the 1850s to escape poverty and overcrowding in east coast cities such as Boston and New York. But lack of wooden fences meant farmers

could not keep cattle off their crops. The invention of barbed wire (made commercially successful by Joseph Glidden in 1874) solved this problem and transformed farming in the Midwest in the process. Prairie grassland was hard to cultivate, but the development of farm machinery, such as John Deere's sodbuster plow, changed this situation. High-yield crops from the Americas, such as peanuts, started making their way to China and East Asia, where, after their

introduction, significant increases in population took place. These crops from the Americas triggered an agricultural revolution in Africa. Corn and sweet potatoes reached the western shores of Africa with slave traders, who introduced them into that continent to provide food for their human cargoes.

Preservation and transport

Equally important to this improvement was that crops could be preserved and distributed efficiently and safely. Although drying and salting had been used for millennia, new preservation techniques were invented during the 18th and 19th centuries. In 1795, French chef Nicolas Appert developed a process for preserving food in airtight bottles after sterilization. This process was patented in England by Peter Durand in 1810, including the provision of using “tin canisters.” The rights to this were bought by engineers Donkin, Hall & Gamble, who set up the world's first canning factory in 1813 in London. Modern refrigeration began in the 1850s when the French inventor Ferdinand Carré pioneered the vapor compression system.

Preserved food made its way around the world on steam railroads and steamships (see pp.292–95). This also enabled agriculture to pursue price and profit globally, rather than focus on a small, local economy, allowing farmers to import specialty seeds, foods, and livestock that suited their terrain.

At last, the population could expand without risk of starvation. A period of high productivity and low food prices meant people did not spend all their money on food. Amid this wealth, however, was the Irish potato famine (see right), which ravaged the Irish population.

The involvement of technology in food production continues in the 21st century, though not without controversy.

FAST-FREEZING

Clarence Birdseye developed advanced techniques for freezing food in the 1920s. The popularity of frozen food increased during World War II 392–405 >>, as it used less packaging than canning.

ADDITIVES

As more people moved to cities, food had to be transported and stay fresh for longer. This was made possible by the introduction of additives to prevent food decay.



FOOD MOUNTAIN

FAMINE

VERSUS PLENTY

The 20th century witnessed increasing concern about commercial and intensive agriculture 462–463 >> which led to **large food surpluses** in developed countries, while the other parts of the world suffered from **famine**.

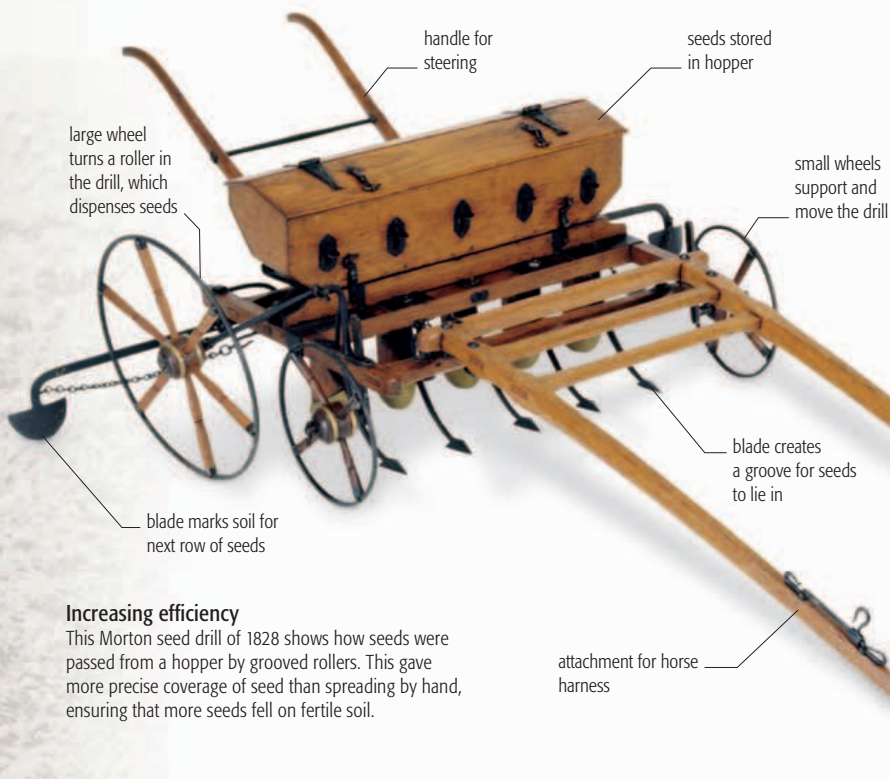
GM FOODS

Since James Watson and Francis Crick broke the genetic code in 1953, the **engineering of genes** has been possible. Genetically modified food products appeared in stores in the 1990s. However, the industry stalled due to misgivings about the safety of GM foods 462–63 >>.

DECISIVE MOMENT

THE IRISH POTATO FAMINE

In 1846, blight—a fungus that damages plants—ruined the potato crop in Europe, leading to starvation for many people in Ireland, where potatoes were a staple food. There was another blight in 1847, accompanied by an outbreak of typhoid. All attempts to deal with the problem failed and Ireland's population was decimated in this, the last peacetime famine in Western Europe. By 1851, the population of Ireland had fallen from 8.5 million to 6.5 million. Many people had died, while others had emigrated, principally to the US.



Increasing efficiency

This Morton seed drill of 1828 shows how seeds were passed from a hopper by grooved rollers. This gave more precise coverage of seed than spreading by hand, ensuring that more seeds fell on fertile soil.